Junxian Li

Basic Information

Name	Junxian Li	**Gender**	Male
Date of Birth	2003-5-1	**Country**	China
Tel	(+86) 19913802609	**Education**	PhD Student, Shanghai Jiao Tong University
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		Links	Home Page; Google Scholar

Self- knowledge

I'm a first-year Ph.D. student in Shanghai Jiao Tong University, major in computer science. My research interests are in MLLMs & UMMs, CV and AI Security.

Education

Xi'an Jiaotong University	Undergraduate	Computer Science	2020.09-2024.06
Shanghai Jiao Tong University	Master → Ph.D.	Computer Science	2024.09-now

School Grades

Ranking (undergrad)：3/191 (all students in my major). Average total score: 92.24

Awards and Group Membership

Awards of Competitions：

Provincial third prize of university students' mathematics competition

Provincial third Prize of English Competition for University Students

Scholarship and School awards：

National Scholarship Award (1st year, Shanghai Jiao Tong University)

First-class scholarship at school level in freshman year **(14 students of 1200 won the scholarship)**

Xi'an Jiaotong University · China Guanggu Scholarship, 2021-2022 **(only I won this in my grade)**

English ability

CET-4 score: 610 **CET-6 score: 601** **CET-6 spoken English score: A**

Intern Experience

Research intern in Tencent Youtu Laboratory	2025.11-2026.03
Research intern in Shanghai Artificial Intelligence Laboratory	2024.03-2025.04

Research experiences and publications

Research Interest: MLLMs & UMMs, Computer Vision, AI security

Junxian Li

ChemVLM: Exploring the power of multimodal large language models in chemistry area (2024-03~2024-08) (First author, AAAI 2025, CCF-A)

IAG: Input-aware Backdoor Attack on VLM-based Visual Grounding (2025-08~2025-12) (First author, CVPR 2026, CCF-A)

Faithful-First Reasoning, Planning, and Acting for Multimodal LLMs (2025-11~2026-01) (First author, ACL 2026, CCF-A)

G^2TR: Generation-Guided Visual Token Reduction for Separate-Encoder Unified Multimodal Models. (First author, NeurIPS 2026, CCF-A)

SlowBA: An efficiency backdoor attack towards VLM-based GUI agents (2026-01~2026-03) (First author, ECCV 2026, CCF-B)

Critic-v: VLM critics help catch VLM errors in multimodal reasoning (2024-09~2024-11) **(co-first author, CVPR 2025, CCF-A)**

Speak-to-Structure: Evaluating LLMs in Open-domain Natural Language-Driven Molecule Generation **(co-first author, KDD 2026, CCF-A)**

S^2 -MLLM: Boosting Spatial Reasoning Capability of MLLMs for 3D Visual Grounding with Structural Guidance. (2025-04~2025-12)

AttnDiff: Attention-based Differential Fingerprinting for Large Language Models. **(NeurIPS 2026, CCF-A)**

TwinPrune: Density-Aware Two-Phase Token Pruning for Vision-Language Models **(NeurIPS 2026, CCF-A)**

delta-Mem: Efficient Online Memory For Large Language Models **(NeurIPS 2026, CCF-A)**

PermuQuant: Lowering Per-Group Quantization Error by Reordering Channels for Diffusion Models **(NeurIPS 2026, CCF-A)**

StreamMeCo: Long-Term Agent Memory Compression for Efficient Streaming Video Understanding (2025-11~2026-01) **(ACL 2026 Findings, CCF-A)**

Reinforcement Learning approaches dealing with missing data for traffic control (2022-07 ~ 2023-01) **(IJCAI 2023, CCF-A)**

Uncertainty-aware Traffic Prediction under Missing Data (2023-03~2023-07) **(ICDM 2023, CCF-B)**

Biology Instructions: A Dataset and Benchmark for Multi-Omics Sequence Understanding Capability of Large Language Models. (EMNLP 2025 Findings, CCF-B)

Lora-chem: modular machine learning for multitask prediction in organic reactions. **(CCS Chemistry, JCR Q1, IF=9.2)**

Teaching MLP more graph information: A three-stage multitask knowledge distillation framework (2023-05~2023-10) Arxiv preprint. (First Author)

Control-R: Towards controllable test-time scaling. Arxiv preprint.

Mol-R1: Towards Explicit Long-CoT Reasoning in Molecule Discovery. Arxiv preprint.

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(Two papers under review of Usenix'Security 2026 and NeurIPS 2026, all first author)

Research Topics

Reinforcement Learning Approaches for graph-based data mining tasks. 2022/07-2023/09
At Xi'an Jiao Tong University, directed by Prof. Bin Shi and Assistant Prof. Hua Wei

Vision Language Models in the scientific Area. 2024/02-2024/09
In Shanghai Artificial Intelligence Laboratory, directed by Prof. Wanli Ouyang and Dr. Dongzhan Zhou

Reasoning for Vision Language Models 2024/10-present
In Shanghai Artificial Intelligence Laboratory, working with Di Zhang, Ph.D. of Fudan University

Efficient MLLM reasoning, MLLM security 2023/10-present.
At Shanghai Jiao Tong University, directed by Prof. Haojin Zhu and Prof. Yulun Zhang

Academic Skills

Proficient in deep learning skills and Python, PyTorch utilization, and deep learning algorithm implementation.
Proficient in VLM fine-tuning and evaluation, especially Qwen, LlaVA, InternVL, etc. (First-author of an InternVL-

based project).
Proficient in data collection and reading literature and technical documents.
Self-motivated, proficient in driving a project